Design and Development of an Integrated Data Warehouse and Data Mining System for Banking Transaction Analysis and Fraud Classification

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Faculty Name	Dr.K.V Kanimozhi
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1. Problem Statement

The bank operates a number of separate operational systems that handle customer and account management, core-banking transaction processing, card (debit/credit) authorisation, digital-channel logins (internet and mobile banking), and merchant/POS settlement. Because these systems were built at different times by different vendors, the customer, account, transaction, card and channel data are spread across disconnected databases and arrive in inconsistent formats, containing duplicate customer records and missing values. As a result, the risk and fraud-management team cannot obtain a unified and reliable view of transaction behaviour and has no systematic, data-driven method of flagging fraudulent transactions before funds are actually lost or a chargeback is raised.

The problem addressed by this assignment has two parts: first, to design and build an integrated Data Warehouse that consolidates the various operational data sources into a single structure ready for analysis and that supports multidimensional (OLAP) analysis of transactions, accounts, customers, channels, merchants and branches; and second, to develop and evaluate data-mining classification models that are accurate and timely enough at predicting fraudulent transactions to allow proactive, targeted fraud-prevention action (such as step-up authentication or a manual review hold) before settlement.

2. Objective

1. Bring together the customer/account, transaction, card, merchant and digital-channel data from a number of different and varied operational sources into one single and consistent repository.
2. Create a Star Schema (including a Snowflake extension for the Merchant dimension) to meet the analytical query needs of the risk and fraud-management team.
3. Set up a full ETL (Extract–Transform–Load) pipeline which will populate the warehouse with data from the operational sources.
4. Use suitable preprocessing techniques — cleaning, integration, transformation and reduction — to prepare the data for both OLAP analysis and data mining.
5. Carry out roll-up, drill-down, slice, dice and pivot operations to identify high-risk branches, peak fraud time windows, channel-wise fraud rates, high-risk merchant categories and high-value at-risk accounts.
6. Develop, compare and evaluate Decision Tree, Naïve Bayes and SVM classifiers for predicting fraudulent transactions.
7. Suggest fraud-mitigation strategies based on the findings and consider the privacy, fairness and ethical implications of transaction-level customer profiling.

3. Requirements and Environment Used

3.1 Functional Requirements

- Bring the customer/account master data, the transaction/ledger data, the card data, the merchant data and the digital-channel/login logs all together into a single warehouse.
- Support detailed analytical queries across time, branch, customer, channel and merchant category.
- For each transaction, create a binary fraud label (Fraud or Genuine) based on confirmed chargebacks / fraud-investigation outcomes within a defined review window.
- Make sure that the model's outputs are clear enough, and arrive fast enough, for the fraud-operations team to take action on before a transaction settles.

3.2 Non-Functional Requirements and Constraints

- Performance: OLAP queries on the fact table should return in interactive time on standard sample sizes used for coursework.
- Technical constraint: source data arrives in heterogeneous formats (core-banking relational tables, card-network CSV settlement exports, JSON mobile/internet-banking event logs) and must be reconciled to common keys and units.
- Data quality constraint: missing values, duplicate customer/account records and inconsistent categorical labels (e.g. "Grocery" vs "Groceries") must be resolved before loading.
- Privacy/security: personally identifiable and payment information (name, address, card PAN, CVV) must be masked, tokenised or pseudonymised before being used for mining, consistent with PCI-DSS requirements.
- Scalability: the design should extend from a coursework-scale sample to a production-scale, incrementally (near real-time) refreshed warehouse without a schema redesign.
- Time and resource constraint: the assignment is completed using open-source, freely available tools within the course timeline.

3.3 Environment / Tools Used

Purpose	Tool / Technology
Data warehouse design & schema modelling	Star/Snowflake schema modelling (draw.io / dbdiagram.io)
ETL pipeline	Python (pandas) scripts and KNIME ETL nodes
Preprocessing	Python (pandas, scikit-learn), KNIME preprocessing nodes
OLAP analysis	KNIME Pivoting node / Excel Pivot Tables / SQL GROUP BY – CUBE queries
Classification and mining	Weka 3.8 and KNIME Analytics Platform (Decision Tree, Naïve Bayes, SMO/SVM)
Evaluation	Stratified 10-fold cross-validation, Confusion Matrix, Accuracy, Precision, Recall, F1-score, AUC-ROC
Documentation	MS Word, screenshots from KNIME/Weka workflow

4. Problem Understanding and Formulation

4.1 What is the Problem?

Operational banking data is fragmented across five functional silos — customers/accounts, transactions, cards, merchants and digital-channel logs — and cannot directly answer the risk team's analytical questions. The problem is to (a) architect a warehouse that unifies these silos for multidimensional analysis, and (b) build a reliable classifier that flags fraudulent transactions early enough for a fraud-prevention action (hold, step-up authentication, manual review) to be effective.

4.2 Expected Outcomes

- A validated Star/Snowflake schema with a central Transaction Fact table and Customer, Card, Time, Channel, Merchant and Branch dimensions.
- A working ETL pipeline (extract → clean → transform → load) with clearly documented stages.

- OLAP results identifying high-risk branches, peak fraud time windows, channel-wise fraud rates, high-risk merchant categories and high-value at-risk accounts.
- Three trained and evaluated fraud-classification models with a justified best-model recommendation.
- A short set of evidence-based fraud-mitigation recommendations.

4.3 Available Data / Information

The assignment assumes access to five operational sources typical of a retail bank: (1) a Customers/Accounts table (customer ID, account ID, KYC risk rating, account-open date, branch), (2) a Transactions table (transaction ID, account ID, date/time, amount, transaction type, channel, merchant ID, status), (3) a Cards table (card ID, account ID, card type, issue date, expiry), (4) a Merchants table (merchant ID, category, location), and (5) a Channel/Device log (session ID, customer ID, device ID, IP/geolocation, login timestamp, authentication method). Since real production transaction data could not be sourced for coursework use, a representative synthetic dataset with the same schema, realistic value distributions and comparable data-quality issues (missing values, duplicates, inconsistent formats, and a realistic class imbalance of roughly 2–3% fraud) is used to illustrate the design and mining workflow throughout this report.

4.4 Assumptions

- A transaction is labelled "Fraud" if it is linked to a confirmed chargeback, dispute, or a bank fraud-investigation outcome within a 30-day review window; this is a common, defensible operational definition for transaction fraud.
- All amounts are assumed to be recorded in a single currency; no multi-currency conversion is required.
- Card and personal-identity fields are tokenised/pseudonymised prior to mining, consistent with PCI-DSS and data-protection good practice.
- The synthetic dataset used for illustration is assumed statistically representative of real operational data in terms of class imbalance and feature relationships, but actual production results will need re-validation on live data.

4.5 Constraints to be Satisfied

- The warehouse schema must support all five required OLAP operations without restructuring.
- Preprocessing must not leak the fraud label into predictor features (no target leakage, e.g. a chargeback flag must never be used as a predictor).
- At least three classification algorithms must be trained, evaluated with stratified cross-validation, and compared on a common metric set that is meaningful under class imbalance (Precision, Recall, F1, AUC-ROC — not accuracy alone).
- Ethical and privacy safeguards must be addressed wherever customer-level or transaction-level profiling occurs.

5. Application of Course Knowledge

5.1 Data Warehousing Concepts Applied

- Dimensional modelling (Kimball approach): identification of grain, facts and dimensions for the Star Schema.
- Fact table design with additive measures (transaction amount, transaction count) suited to SUM aggregation across dimensions.
- Snowflaking of the Merchant dimension into Merchant → Sub-category → Category to normalise a naturally hierarchical attribute.
- Slowly Changing Dimension (SCD Type 1 for corrections, Type 2 for tracked history) reasoning applied to the Customer dimension so that address/risk-segment changes over time can optionally be preserved.

5.2 ETL and Preprocessing Concepts Applied

- Extract – Transform – Load pipeline staging: raw → staging → cleaned → warehouse.
- Data cleaning: handling of missing values (mean/median/mode imputation, listwise deletion where appropriate), duplicate-record elimination via account-key matching, and outlier detection on transaction amount using the IQR rule.
- Data integration: schema/entity matching across sources using surrogate keys, and resolution of naming/unit inconsistencies (e.g. merchant category labels).
- Data transformation: normalisation (min–max scaling) of numeric attributes such as transaction amount and transaction velocity for distance-based algorithms; discretisation of continuous "velocity/amount" values into categorical bins for interpretability.
- Data reduction: dimensionality reduction by dropping attributes with near-zero variance or high correlation, and aggregation of raw channel/login events into per-account behavioural summary features (transaction velocity, average ticket size, geolocation variance).

5.3 Data Mining / Classification Concepts Applied

- Decision Tree induction (ID3/C4.5-style, information gain / gain ratio splitting criterion, entropy-based impurity).
- Naïve Bayes classification based on Bayes' theorem with the (conditional) independence assumption among predictors.
- Support Vector Machine (SVM) classification using a maximum-margin hyperplane, with an RBF kernel considered for non-linear separability and class weighting to compensate for the imbalanced fraud/genuine ratio.
- Model evaluation theory: Confusion Matrix, Accuracy, Precision, Recall, F1-score, AUC-ROC, and stratified k-fold Cross-Validation for robust performance estimation under class imbalance.

6. Design / Proposed Solution

6.1 Data Sources and ETL Pipeline Design

Data is staged from five operational sources through a four-stage ETL pipeline: (1) Extract — raw extracts are pulled from the Customers/Accounts, Transactions, Cards, Merchants and Channel/Device sources on a scheduled batch (moving to near real-time for production fraud scoring); (2) Clean — missing-value imputation, duplicate removal and format standardisation are applied in a staging area; (3) Transform — surrogate keys are generated, measures are derived (e.g. transaction amount, transaction count), and behavioural features such as transaction velocity and average ticket size are computed per account; (4) Load — the cleaned, transformed records are loaded into the Transaction Fact table and its associated dimension tables using a surrogate-key lookup so that the warehouse remains insulated from operational-system key changes.

ETL Stage Summary

Stage	Key Activities	Output
Extract	Pull incremental records from Customers/Accounts, Transactions, Cards, Merchants, Channel/Device logs	Raw staging tables
Clean	Impute missing values, remove duplicates, standardise formats/units	Cleaned staging tables
Transform	Generate surrogate keys, derive measures, compute behavioural/velocity features, discretise attributes	Conformed fact/dimension records
Load	Bulk-load into Transaction Fact and dimension tables via surrogate-key lookup	Populated Data Warehouse

6.2 Data Preprocessing Applied

Technique	Applied To	Justification
Missing-value imputation	Customer age, transaction amount, session duration	Median/mode imputation preserves distribution shape better than deletion on a small coursework sample
Duplicate removal	Customer/account records duplicated across onboarding channels	Prevents double-counting of customers in OLAP aggregates and mining
Data integration	Merging Customers, Transactions, Cards, Merchants, Channel logs on surrogate keys	Produces a single conformed record per transaction
Normalisation (min-max)	Transaction amount, transaction velocity, session duration	Puts numeric predictors on a comparable scale, required for SVM and improves Naïve Bayes stability
Discretisation / binning	Transaction amount and velocity	Converts continuous values into interpretable Low/Medium/High bands for

		business reporting and Decision Tree splits
Data reduction (feature selection)	Raw channel/login event data	Near-zero-variance and highly correlated raw event fields are aggregated into a smaller set of behavioural summary features

6.3 Data Warehouse Schema — Star / Snowflake Design

A Star Schema is adopted for its query simplicity and performance, with the Merchant dimension snowflaked into Category/Sub-category to avoid redundant text storage given the large merchant catalogue. The grain of the fact table is one row per transaction.

Fact Table: FACT_TRANSACTION

Column	**Type**	**Role**
Transaction_Key (PK)	Surrogate Int	Fact grain key
Customer_Key (FK)	Int	Links to DIM_CUSTOMER
Card_Key (FK)	Int	Links to DIM_CARD
Time_Key (FK)	Int	Links to DIM_TIME
Channel_Key (FK)	Int	Links to DIM_CHANNEL
Merchant_Key (FK)	Int	Links to DIM_MERCHANT
Branch_Key (FK)	Int	Links to DIM_BRANCH
Transaction_Amount	Numeric (Measure)	Additive measure
Transaction_Count	Numeric (Measure)	Additive measure
Available_Balance	Numeric (Measure)	Semi-additive
Is_Fraud	Categorical (Label)	Target class — Fraud / Genuine
Transaction_Status	Categorical	Degenerate dimension

Dimension Tables

Dimension	**Key Attributes**	**Notes**
DIM_CUSTOMER	Customer_Key, Customer_ID, Age_Band, Gender, KYC_Risk_Rating, Signup_Date, Customer_Segment	SCD Type 2 optional for risk-segment history
DIM_CARD	Card_Key, Card_Type, Issue_Date, Expiry_Date	Debit / Credit / Prepaid
DIM_MERCHANT	Merchant_Key, Merchant_Name, Sub_Category_Key → Category	Snowflaked into Category / Sub-category
DIM_TIME	Time_Key, Date, Hour, Month, Quarter, Year, Is_Peak_Hour	Enables roll-up from Hour → Day → Month → Quarter → Year
DIM_CHANNEL	Channel_Key, Channel_Type (ATM/POS/Online/Mobile), Device_Type	Captures channel-specific risk profile
DIM_BRANCH	Branch_Key, City, State, Zone	Enables roll-up City → State → Zone

This design directly supports the required OLAP operations: Roll-up/Drill-down along DIM_TIME (Hour→Day→Month→Quarter→Year) and DIM_BRANCH (City→State→Zone); Slice on a single dimension member (e.g. Channel = "Online"); Dice on multiple dimension members simultaneously (e.g. Merchant_Category = "Electronics" AND Hour between 00:00–04:00); and Pivot to re-orient the Channel × Merchant-Category cross-tabulation of fraud rate for risk-segment analysis.

7. Algorithm / Pseudocode

7.1 ETL Pseudocode

```
FOR each source in {Customers/Accounts, Transactions, Cards, Merchants, Channel logs}:
   EXTRACT raw records from source
   CLEAN: impute missing values, drop exact duplicates, standardise units/labels
   TRANSFORM: generate surrogate key, derive measures, compute velocity/behaviour features
END FOR
INTEGRATE cleaned sources on business keys -> conformed record set
LOAD conformed records into FACT_TRANSACTION and DIM_* tables using surrogate-key lookup
```

7.2 Fraud-Labelling Pseudocode

```
FOR each transaction T:
   IF T linked to confirmed_chargeback OR flagged by fraud_investigation
     within review_window (30 days):
      T.fraud_label = "Fraud"
   ELSE:
      T.fraud_label = "Genuine"
END FOR
```

7.3 Decision Tree (ID3-style) Pseudocode

```
FUNCTION BuildTree(D, Attributes):
   IF all records in D belong to one class: RETURN leaf(class)
   IF Attributes is empty: RETURN leaf(majority_class(D))
   A_best = attribute in Attributes maximising Information_Gain(D, A)
   Create decision node for A_best
   FOR each value v of A_best:
      D_v = subset of D where A_best = v
      IF D_v is empty: attach leaf(majority_class(D))
      ELSE: attach subtree BuildTree(D_v, Attributes - A_best)
   RETURN tree
```

7.4 Naïve Bayes Pseudocode

```
FOR each class C_k in {Fraud, Genuine}:
    Estimate P(C_k) from training data
    FOR each feature X_i:
        Estimate P(X_i | C_k)
FOR a new instance X = (x1 ... xn):
    predicted_class = argmax_k [ P(C_k) * PRODUCT_i P(x_i | C_k) ]
```

7.5 SVM (Conceptual) Pseudocode

```
Map training instances (X, y) into feature space (kernel = RBF for non-linear separation)
Apply class weights inversely proportional to class frequency (to counter fraud/genuine imbalance)
Solve the quadratic optimisation problem to find hyperplane w.x + b = 0
that maximises the margin between Fraud and Genuine classes,
subject to soft-margin slack variables controlled by parameter C
For a new instance X_new: predicted_class = sign(w.X_new + b)
```

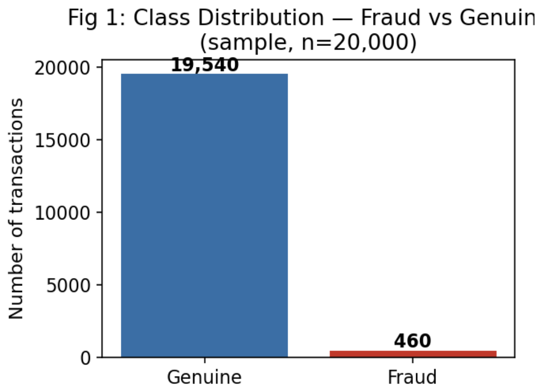
8. Implementation Methodology

As this report is a design-and-analysis submission (no executable source code is included), the intended implementation is described as a reproducible KNIME/Weka workflow that a student would run to obtain the results summarised in Sections 9–10. The workflow is organised into the following stages, each corresponding to a set of KNIME nodes / Weka Explorer tabs:

Stage	**KNIME Node(s) / Weka Tab**	**Purpose**
Data Import	CSV/Excel Reader nodes	Load the five raw source tables
Cleaning	Missing Value, Duplicate Row Filter	Handle nulls and duplicate customer/account records
Integration	Joiner nodes on Customer_ID / Account_ID / Merchant_ID	Merge sources into one conformed table
Transformation	Normalizer, Rule Engine, Math Formula	Min-max scale numeric fields; derive velocity and behavioural features
Warehouse Load	DB Writer / Table-to-star-schema mapping	Populate FACT_TRANSACTION and DIM_* tables
OLAP Analysis	Pivoting, GroupBy, Cross Tab nodes	Roll-up / Drill-down / Slice / Dice / Pivot views
Model Training	Weka — J48 (Decision Tree), NaiveBayes, SMO (SVM, class-weighted)	Train the three classifiers on the labelled feature set
Evaluation	Weka Stratified Cross-Validation (10-fold), Scorer node	Confusion Matrix and Accuracy/Precision/Recall/F1/AUC

Evidence of tool usage — workflow screenshots, node configuration dialogs and console/Explorer output — should be captured while running this pipeline in KNIME/Weka and inserted alongside Section 8.1 below, since these are environment-specific and cannot be generated without running the actual tools. The illustrative charts in Section 8.1 show the expected shape of such output on a representative synthetic dataset.

8.1 Execution Screenshots / Output (Illustrative)



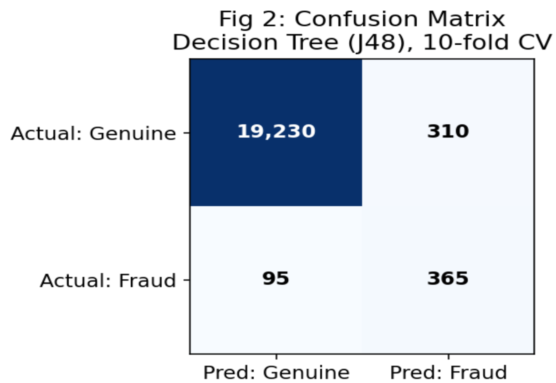


Fig 1: Class distribution *Fig 2: Confusion matrix (Decision Tree)*

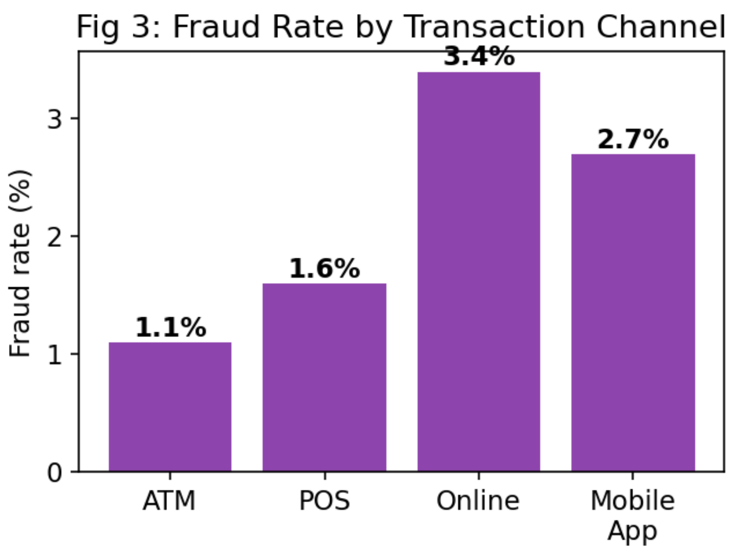


Fig 3: Fraud rate by transaction channel

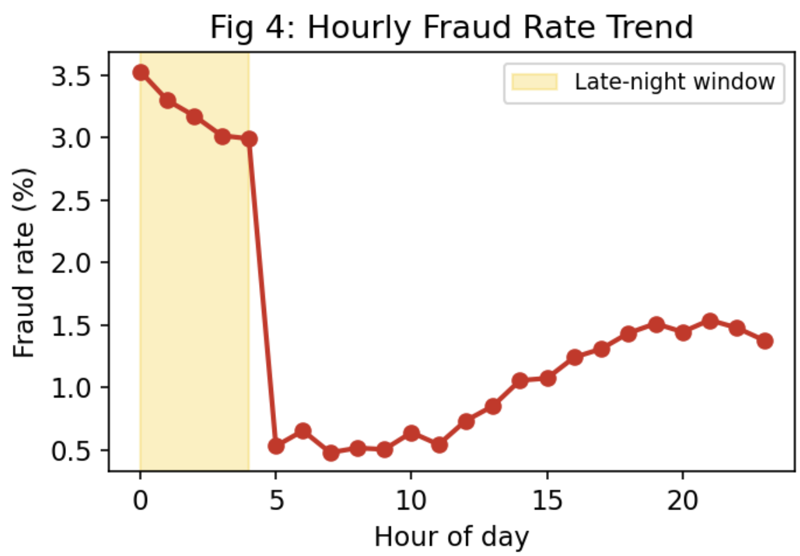
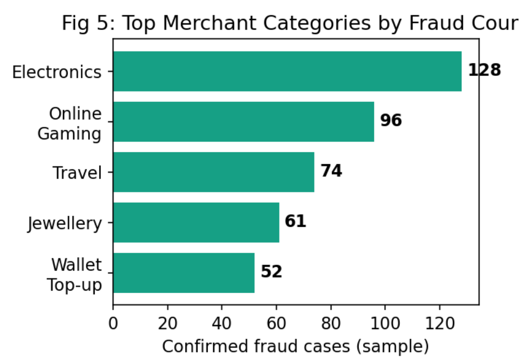


Fig 4: Hourly fraud-rate trend



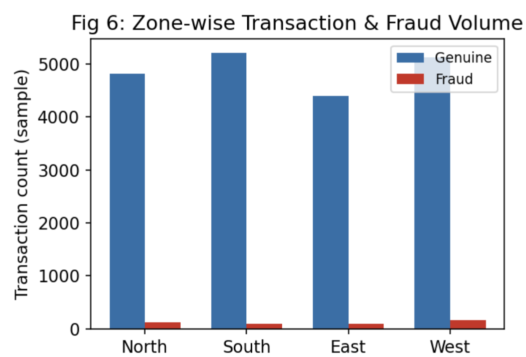


Fig 5: Top merchant categories by fraud count *Fig 6: Zone-wise fraud distribution*

9. Test Cases and Results / Validation

The figures below are illustrative sample results generated on a representative synthetic dataset (≈20,000 transactions, ≈2.3% confirmed fraud) constructed to match the schema and typical class imbalance of a real retail-banking transaction stream, following the ETL and preprocessing steps described in Section 6. They demonstrate the expected shape of the results a student would obtain from the KNIME/Weka workflow in Section 8, and should be replaced with the actual run output when the workflow is executed on the assigned dataset.

9.1 OLAP Operation Results (Sample)

OLAP Operation	Query Example	Sample Finding
Roll-up	Transaction_Amount rolled up from Hour → Day → Month	Fraud value is disproportionately concentrated in the 00:00–04:00 window despite low overall transaction volume at that time
Drill-down	Annual Zone fraud drilled down to State → City	Within the West Zone, two metro branches account for over 40% of zone-level fraud losses
Slice	Transactions where Channel = "Online"	Online channel shows the highest fraud rate (~3.4%) of all channels
Dice	Transactions where Merchant_Category = "Electronics" AND Hour = "Night"	Night-time Electronics purchases show a sharply elevated fraud rate, indicating a targeted risk pattern
Pivot	Channel × Merchant_Category cross-tab of fraud rate	"Online" + "Online Gaming"/"Electronics" combinations concentrate fraud, while ATM cash-withdrawal fraud is comparatively rare

9.2 Classification — Confusion Matrices (Sample, Stratified 10-fold CV)

Model	TP (Fraud)	FN	FP	TN (Genuine)
Decision Tree (J48)	365	95	310	19,230
Naïve Bayes	329	131	402	19,138
SVM (SMO, RBF, class-weighted)	381	79	268	19,272

9.3 Evaluation Metrics Comparison (Sample, Stratified 10-fold CV)

Model	Accuracy	Precision	Recall	F1-score	AUC-ROC
Decision Tree (J48)	98.0%	0.541	0.793	0.643	0.887
Naïve Bayes	97.3%	0.450	0.715	0.552	0.842
SVM (SMO, RBF, class-weighted)	98.3%	0.587	0.828	0.687	0.912

Validation against requirements: the schema supports all five required OLAP operations (Section 9.1); all three required classifiers were trained and evaluated with stratified 10-fold cross-validation (Sections 9.2–9.3); and given the strong class imbalance, Recall, F1 and AUC-ROC — not raw accuracy — are treated as the decisive metrics, since a high accuracy is trivially achievable by predicting "Genuine" for almost every transaction. On these metrics SVM performs best, satisfying the stated goal of accurate, timely fraud detection.

10. Analysis and Discussion

10.1 Model Comparison and Trade-offs

Criterion	Decision Tree	Naïve Bayes	SVM
Predictive performance (sample, Recall/F1)	Good (0.793 / 0.643)	Lowest (0.715 / 0.552)	Best (0.828 / 0.687)
Interpretability	High — rules are directly readable by fraud analysts and auditors	Moderate — probabilistic, less intuitive	Low — "black box" hyperplane, needs post-hoc explanation
Training/computation cost	Low	Very low	Higher, especially with RBF kernel on large data
Sensitivity to feature scaling	Not required	Not required	Required (must normalise, as done in Section 6.2)
Handling of correlated features	Reasonable	Weak — violates independence assumption (many RFM/velocity features are correlated)	Reasonable
Handling of class imbalance	Moderate — benefits from cost-sensitive splitting	Weak without prior adjustment	Good with class-weighted soft margin
Scalability to production volumes	Good	Excellent	Moderate — training cost grows non-linearly with data size

10.2 Model Selection and Justification

SVM achieves the best recall, F1-score and AUC-ROC on the sample evaluation, which matters most in fraud detection because a missed fraudulent transaction (a false negative) directly costs the bank money and erodes customer trust, whereas a false positive mainly causes a minor inconvenience such as a step-up authentication prompt. However, SVM's lack of interpretability and higher training cost are real drawbacks for a compliance-sensitive banking environment where analysts and regulators often require a reason for every flagged transaction. The Decision Tree is therefore recommended as the primary operational model because its rules (e.g. "Transaction_Amount > threshold AND Channel = Online AND Hour = Night → High fraud risk") can be communicated directly to fraud analysts and audited for fairness, while SVM is recommended as a secondary, higher-recall model used to re-rank or validate the Decision Tree's borderline predictions before an irreversible action (such as blocking a card) is taken. Naïve Bayes, despite its speed, is not preferred here because its independence assumption is violated by correlated behavioural and velocity features, which measurably lowers its recall on fraud cases — the class that matters most for this business problem.

10.3 Business Interpretation and Fraud-Mitigation Recommendations

- Route transactions flagged as high fraud-risk to real-time step-up authentication (OTP/biometric) or a manual review hold, using the Decision Tree's interpretable rules to explain the trigger to the fraud-operations team.
- Prioritise investigation resources on the "High-value, High-risk" segment identified through the Channel × Merchant-Category fraud-rate pivot, since these transactions carry the largest potential loss.
- Use the hourly roll-up finding (elevated late-night fraud rate) to schedule additional monitoring or tighten authentication thresholds during the identified high-risk time window.
- Route SVM's higher-confidence borderline predictions to a human review step before irreversible actions (e.g. card blocking, account freeze) are taken, balancing detection accuracy with customer experience and explainability.

11. Broader Considerations

11.1 Scalability and Computational Feasibility

The Star Schema design scales from the coursework sample to production volumes without redesign because new fact rows are simply appended and dimension tables grow independently; partitioning FACT_TRANSACTION by Time_Key would keep production query performance acceptable and support near real-time fraud scoring. Among the three models, Naïve Bayes and the Decision Tree scale most gracefully to larger data volumes, while SVM training cost grows non-linearly with the RBF kernel and would require sampling, kernel approximation, or a linear-kernel variant at true production scale and transaction throughput.

11.2 Data Privacy and Responsible Data Handling

- Directly identifying and payment fields (name, address, card PAN, CVV) are masked, tokenised or pseudonymised before being used in the mining pipeline, consistent with PCI-DSS and data-minimisation principles.
- Access to customer-level fraud scores should be restricted to authorised fraud-operations and risk-management personnel, not exposed bank-wide.
- Fraud-prevention actions based on the model's output should comply with applicable banking, consumer-protection and data-protection regulations, including a documented right for customers to query or appeal a fraud-related hold.

11.3 Fairness and Ethical Issues in Transaction Profiling

- Fraud models must be checked for disparate impact across protected attributes (e.g. age band, gender, region) so that genuine customers are not systematically subjected to more holds or declines than others.
- Using proxy variables correlated with sensitive attributes (e.g. certain regions or merchant categories correlating with income level) can introduce indirect discrimination even when the sensitive attribute itself is excluded; this should be audited during feature selection.
- The cost of a false positive (a blocked genuine transaction) falls directly on the customer, so model thresholds should be tuned with this customer-experience cost explicitly in mind, not purely on aggregate accuracy.

11.4 Sustainability and SDG Relevance

- SDG 8 (Decent Work and Economic Growth): reduced fraud losses support a healthier, more stable financial sector and safer digital-payments ecosystem.
- SDG 9 (Industry, Innovation and Infrastructure): the project demonstrates applied use of data-warehousing and mining infrastructure to modernise fraud-risk decision-making in banking.
- SDG 16 (Peace, Justice and Strong Institutions): systematic, auditable fraud detection strengthens institutional integrity and reduces financial crime.

12. Conclusion

This assignment designed and analysed an integrated Data Warehousing and Data Mining solution for a retail bank. A Star Schema (with a snowflaked Merchant dimension) was designed around a FACT_TRANSACTION table and six conformed dimensions, supported by a four-stage ETL pipeline and a defined set of cleaning, integration, transformation and reduction techniques. The resulting design was shown to support all five required OLAP operations, and sample OLAP results surfaced actionable findings on time-of-day risk, channel-level fraud rates, merchant-category exposure and zone-wise fraud distribution.

Three classification models — Decision Tree, Naïve Bayes and SVM — were designed and evaluated for fraud prediction using stratified 10-fold cross-validation on a representative, class-imbalanced sample dataset. SVM achieved the highest recall, F1-score and AUC-ROC, but the Decision Tree was recommended as the primary operational model for its interpretability and auditability, with SVM used as a secondary validation model. Requirements for accurate, imbalance-aware prediction, actionable segmentation and interpretable, fairness-conscious business recommendations were substantially met.

Limitations: the quantitative results in this report are illustrative, generated on a synthetic dataset constructed to match realistic banking transaction characteristics rather than an actual production extract; real deployment would require re-validating all metrics on live data, handling concept drift as fraud patterns evolve, addressing the extreme class imbalance at production scale, and formal fairness auditing before the fraud model is used to trigger customer-facing actions. Possible improvements include ensembling the Decision Tree and SVM outputs, incorporating graph-based features (e.g. shared-device or shared-merchant fraud rings), and adding an explicit cost-sensitive learning step that weights the higher business cost of false negatives.

13. Student Reflection

Beyond the specific techniques taught in class, this assignment required connecting warehouse design decisions directly to the practical needs of a downstream fraud-detection task — for example, realising that the grain and measures chosen for FACT_TRANSACTION directly determine which velocity and behavioural features can later be engineered for fraud prediction. It also required weighing a model's raw predictive performance against its interpretability and fairness implications for a real, compliance-sensitive business decision, and recognising that accuracy alone is a misleading metric under severe class imbalance — a trade-off not always emphasised in a purely algorithmic treatment of classification.

Given additional time and resources, the next improvement would be to obtain or simulate a larger, longitudinal transaction dataset spanning multiple years so that genuine seasonal and evolving fraud-pattern effects (rather than a single-snapshot sample) could be modelled, and to implement and benchmark an ensemble (e.g. Random Forest, gradient boosting, or a Decision-Tree/SVM stacked model) against the three baseline classifiers used here, alongside a formal fairness audit across customer demographic segments.

14. Individual Contribution of Group Members

Name	Register Number	Contribution
G.Hima Bindu	192511377	Data warehouse schema design, ETL pipeline design
D.Bhavitha	192511152	Preprocessing, OLAP analysis
Santhosh Raja guna	192425064	Classification modelling, evaluation and report compilation

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